

Questão

① $f(t) = t - \pi$ no intervalo $[0, 2\pi]$

$$f(t) = f(t + 2n\pi), \quad n \text{ é inteiro}$$

1- $f(t) = f(t + 2n\pi)$ indica que a função é periódica.

Se a função é periódica, então podemos calcular os coeficientes da série de Fourier. $\hookrightarrow$ e obedece as condições de Dirichlet

$$f(t) = f(t + T) = f(t + 2\pi n)$$

Para o menor valor positivo $(n=1)$, $T = 2\pi$

2 - Cálculo da série de Fourier

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi k \cdot t/T}$$

$$X_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-i2\pi k \cdot t/T} dt = \frac{1}{T} \int_0^T x(t) e^{-i2\pi k \cdot t/T} dt$$

$\hookrightarrow$ Coeficiente da série de Fourier

$$x(t) = f(t) = t - \pi$$

$$X_k = \frac{1}{T} \int_0^{2\pi} (t - \pi) \cdot e^{-i2\pi k \cdot t/T} dt$$

$$X_k = \frac{1}{T} \left\{ \underbrace{\int_0^{2\pi} t \cdot e^{-i2\pi k \cdot t/T} dt}_{(1)} - \pi \cdot \underbrace{\int_0^{2\pi} e^{-i2\pi k \cdot t/T} dt}_{(2)} \right\}$$

① Integração por partes

$$T=2\pi \quad \frac{-2i\pi k}{T} = \frac{-2i\pi k}{2\pi} = -ik$$

$$\int_0^{2\pi} t \cdot e^{-ikt} \Rightarrow \int u \cdot v = u \cdot v - \int v \cdot du$$

$$u=t \quad du=dt \quad dv=e^{-ikt} \quad v=\frac{e^{-ikt}}{-ik}$$

$$\frac{t \cdot e^{-ikt}}{-ik} \Big|_0^{2\pi} - \frac{1}{-ik} \cdot \int_0^{2\pi} e^{-ikt}$$

$$\frac{1}{-ik} \left\{ 2\pi \cdot e^{-ik2\pi} - \frac{1}{-ik} \left[e^{-ikt} \right]_0^{2\pi} \right\}$$

$$\frac{1}{-ik} \left\{ 2\pi \cdot e^{-ik2\pi} + \frac{1}{ik} \left[e^{-ik2\pi} - 1 \right] \right\}$$

$$\frac{1}{-ik} \left\{ e^{-ik2\pi} \left(2\pi + \frac{1}{ik} \right) - \frac{1}{ik} \right\}$$

$$ik \cdot -ik = k^2$$

$$\left\{ e^{-ik2\pi} \left(\frac{2\pi}{-ik} + \frac{1}{k^2} \right) - \frac{1}{k^2} \right\}$$

$$e^{-ik2\pi} \cdot \left(\frac{-2\pi \cdot ik + 1}{-ik \cdot ik} \right) - \frac{1}{k^2} = e^{-ik2\pi} \left(\frac{2\pi ik + 1}{k^2} \right) - \frac{1}{k^2}$$

$$= \frac{e^{-ik2\pi} \cdot (2\pi i k + 1) - 1}{k^2} = \left(\frac{2\pi i k + 1}{k^2} \right) - 1 = \frac{2\pi i k}{k^2}$$

$$= \frac{2\pi i}{k}$$

$$e^{-ik2\pi} = \cos \underbrace{(k2\pi)}_0 - i \sin \underbrace{(k2\pi)}_0$$

$$\textcircled{2} \quad \pi \cdot \int_0^{2\pi} e^{-2i\pi k \cdot t/T} dt = \pi \cdot \int_0^{2\pi} e^{-ik t} dt$$

$$T=2\pi \quad \frac{-2i\pi k}{T} = \frac{-2i\pi k}{2\pi} = -ik$$

$$= \pi \cdot \frac{1}{-ik} \left[e^{-ik t} \right]_0^{2\pi} = \frac{\pi i}{k} [e^{-ik2\pi} - 1]$$

$$e^{-ik2\pi} = \cos \underbrace{(k2\pi)}_0 - i \sin \underbrace{(k2\pi)}_0$$

$$= \frac{\pi i}{k} [1 - 1] = 0$$

Integrando ① e ②

$$X_k = \frac{1}{T} \left\{ \int_0^{2\pi} \underset{\textcircled{1}}{1} e^{-2i\pi k \cdot t/T} dt - \pi \cdot \int_0^{2\pi} \underset{\textcircled{2}}{e^{-2i\pi k \cdot t/T}} dt \right\}$$

$$X_k = \frac{1}{2\pi} \cdot \left(\frac{2\pi i}{k} - 0 \right) = \frac{i}{k}$$

$$\frac{1}{T} \int_0^T x(t) e^{-i 2\pi k t/T} dt$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{i 2\pi k t/T}$$

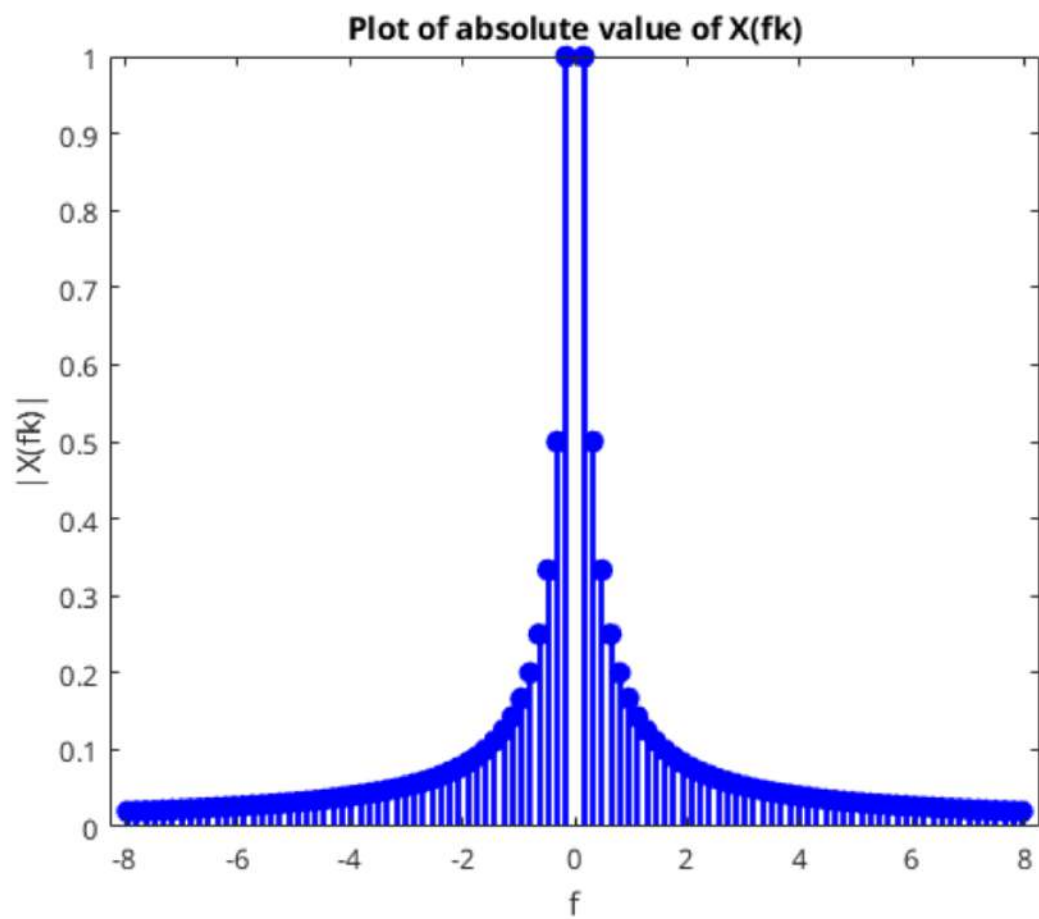
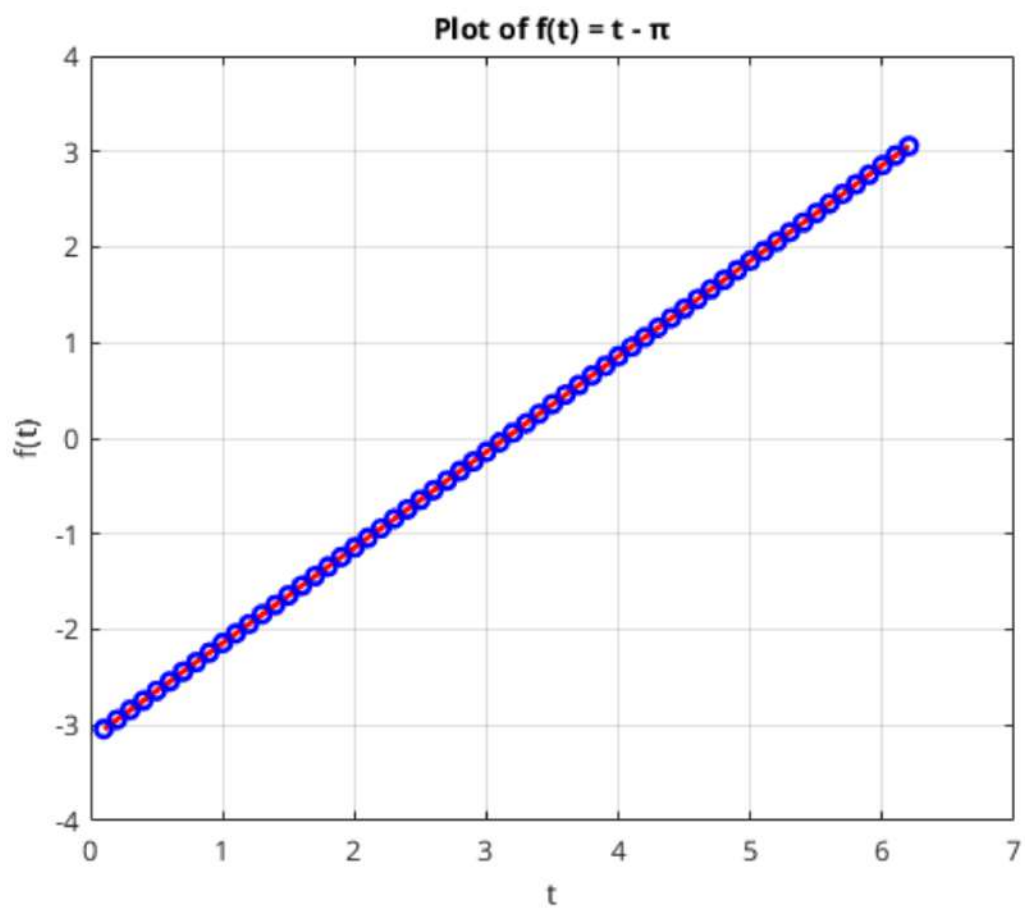
$$X_k = \frac{1}{T} \int_0^T x(t) e^{-i 2\pi k t/T} dt$$

$$T = 2\pi$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{i k t} = \sum_{k=-\infty}^{\infty} \frac{i}{k} \cdot e^{i k t}, k \neq 0$$

$$= i \cdot \sum_{k=-\infty}^{\infty} \frac{e^{i k t}}{k}$$

$$|X_k| = \sqrt{\left(\frac{1}{k}\right)^2} = \frac{1}{k} \sqrt{1} = \frac{1}{k}$$



Questão 2

1- $f(t) = \exp(it) = e^{it}$ intervalo $0 < 2\pi$

$f(t) = f(t + 2n\pi)$, função periódica

$$f(t) = f(t + T) = f(t + 2n\pi)$$

$T = 2n\pi$, e para n inteiro e igual a 1
 $T = 2\pi$

2- Calcule a série de Fourier, supondo que as condições de Dirichlet foram decididas.

$$x(t) = \sum_{k=-\infty}^{\infty} \lambda_k e^{i2\pi k t / T}$$

$$\lambda_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi k t / T} dt$$

Resolvendo a Integral

$$\int_0^{2\pi} e^{t} \cdot e^{-i2\pi k t / T} dt = \int_0^{2\pi} e^{t} \cdot e^{-i2\pi k t / 2\pi} dt$$

$$T = 2\pi$$

$$\int_0^{2\pi} e^{t} \cdot e^{-ik t} dt = \int_0^{2\pi} e^{t(1-ik)} dt = \frac{1}{1-ik} \left[e^{t(1-ik)} \right]_0^{2\pi}$$

$$= \frac{1}{1-ik} \left[e^{2\pi(1-ik)} - 1 \right]$$

$$X_k = \frac{1}{2\pi(1-ik)} \left[e^{2\pi(1-ik)} - 1 \right] = \frac{(e^{2\pi(1-ik)} - 1)}{2\pi(1-ik)}$$

$$e^{2\pi - 2\pi i k} \int^0 = \frac{e^{2\pi}}{e^{2\pi i k}} = e^{2\pi}$$

$$e^{2\pi i k} = \underbrace{\cos(2\pi k)}_1 + i \underbrace{\sin(2\pi k)}_0$$

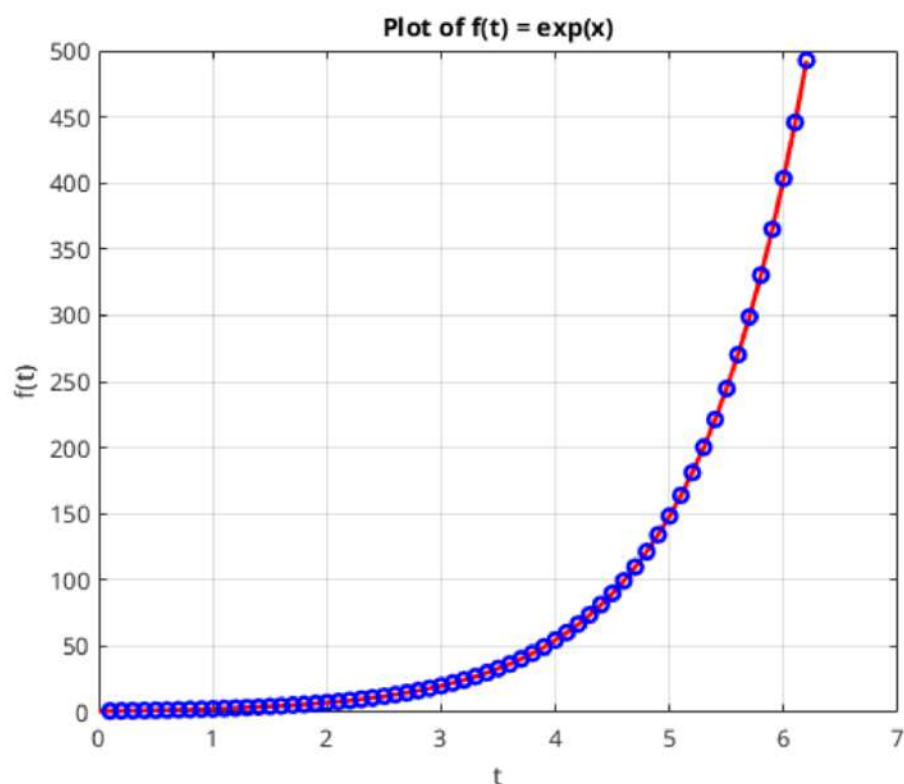
$$X_k = \frac{1}{2\pi} \frac{(e^{2\pi} - 1)}{(1-ik)} \quad X(t) = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi} \frac{(e^{2\pi} - 1)}{(1-ik)} e^{i2\pi k t / T}$$

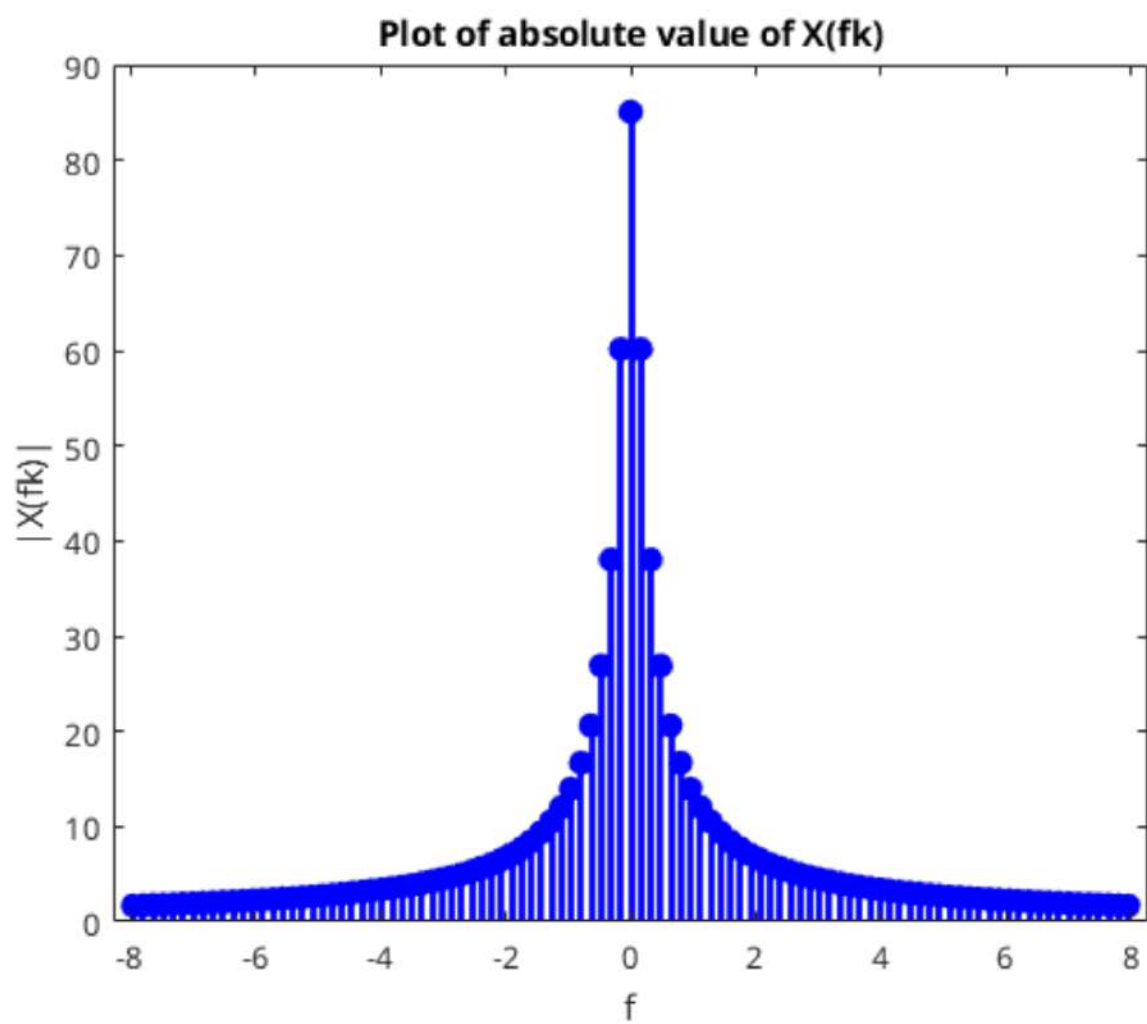
$$T = 2\pi \quad X(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \frac{(e^{2\pi} - 1)}{(1-ik)} e^{ik t}$$

$$|x_k| = \left| \frac{(e^{2\pi} - 1)}{2\pi (i - ik)} \right| = \frac{|(e^{2\pi} - 1)|}{|2\pi (i - ik)|}$$

$$= \frac{|e^{2\pi} - 1|}{|2\pi i - 2\pi k|} = \frac{e^{2\pi} - 1}{\sqrt{(2\pi)^2 + (2\pi)^2}} = \frac{e^{2\pi} - 1}{\sqrt{2\pi^2(k^2 + 1)}}$$

$$= \frac{e^{2\pi} - 1}{2\pi \sqrt{1 + k^2}}$$





Questão 1.3

$$f(t) = e^{2\cos(t)} \cdot \cos(2\sin(t))$$

1- Provar que é uma função periódica

Se $f(t) = f(t+T)$ para todo t pertencente ao domínio de f , então $f(t)$ é uma função periódica.

$$e^{2\cos(t)} \cdot \cos(2\sin(t)) = e^{2\cos(t+T)} \cdot \cos(2\sin(t+T))$$

$$A - \cos(t) = \cos(t+T)$$

$$e B - \cos(2\sin(t)) = \cos(2\sin(t+T))$$

$$A - \cos(t) = \cos(t+T)$$

$$\cos(t) = \cos(t) \cdot \cos(T) - \sin(t) \cdot \sin(T)$$

$$\text{Se } T = 2\pi, \cos(2\pi) = 1, \sin(2\pi) = 0$$

$$\cos(t) = \cos(t) \cdot 1 = \cos(t)$$

$$B - \cos(2\sin(t)) = \cos(2\sin(t+T))$$

$$\cos^{-1} \cos(2\sin(t)) = \cos^{-1} \cos(2\sin(t+T))$$

$$2\sin(t) = 2\sin(t+T)$$

$$\sin(t) = \sin(t+T)$$

$$\sin(t) = \sin(t) \cdot \cos(T) + \sin(T) \cdot \cos(t)$$

$$T = 2\pi \quad \cos(2\pi) = 1 \quad \text{e} \quad \sin(2\pi) = 0$$

$$\text{Logo } \sin(t) = \sin(t)$$

Como

$$A - \cos(t) = \cos(t+T)$$

$$\text{e } B - \cos(2\sin(t)) = \cos(2\sin(t+T))$$

São válidas para um mesmo período $T = 2\pi$

A relação acima é verdadeira

$$e^{2\cos(t)} \cdot \cos(2\sin(t)) = e^{2\cos(t+T)} \cdot \cos(2\sin(t+T))$$

e a função é periódica.

2- Provar que a expansão em série de Fourier

$$\text{é igual a } 1 + \sum_{n=1}^{\infty} \frac{2^n}{n!} \cos(nt)$$

Vamos utilizar o intervalo de 0 a 2π .

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i 2\pi k t / T}$$

$$X_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i 2\pi k t / T} dt$$

$$T = 2\pi$$

$$X_k = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot \cos(2 \sin(t)) \cdot e^{-i \sqrt{\pi} k t / 2\pi} dt$$

$$X_k = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot \cos(2 \sin(t)) \cdot e^{-i k t} dt$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\cos(2 \sin(t)) = \frac{e^{i \cdot 2 \sin(t)} + e^{-i \cdot 2 \sin(t)}}{2}$$

$$\frac{1}{4\pi} \cdot \int_0^{2\pi} e^{2(\cos t)} \cdot e^{-i k t} \cdot (e^{i 2 \sin(t)} + e^{-i 2 \sin(t)}) dt$$

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

$$e^{2 \cos(t)} = e^{2 \cdot \frac{(e^{it} + e^{-it})}{2}} = e^{e^{it} + e^{-it}}$$

$$e^{i2 \sin(t)} = e^{i2 \cdot \frac{(e^{it} - e^{-it})}{2i}} = e^{e^{it} - e^{-it}}$$

$$e^{-i2 \sin(t)} = e^{-i2 \cdot \frac{(e^{it} - e^{-it})}{2i}} = e^{e^{-it} - e^{it}}$$

$$e^{it} = A \quad e^{-it} = B$$

$$\frac{1}{4\pi} \int_0^{2\pi} e^{A+B} \cdot e^{-ikA} \cdot (e^{A-B} + e^{B-A}) dt$$

$$\frac{1}{4\pi} \left\{ \int_0^{2\pi} e^{A+B+A-B-ikA} dt + \int_0^{2\pi} e^{A+B+B-A-ikA} dt \right\}$$

$$\frac{1}{4\pi} \cdot \left\{ \int_0^{2\pi} e^{2A-ikA} dt + \int_0^{2\pi} e^{2B-ikA} dt \right\}$$

$$e^{-ikA} \cdot (e^{2A} + e^{2B})$$

$$e^{it} = \cos(t) + i \sin(t)$$

$$e^{-it} = \cos(t) - i \sin(t)$$

$$e^{2it} = e^{i(\cos(t) + i \sin(t))} + e^{i(\cos(t) - i \sin(t))}$$

$$e^{2i \cos(t)} \left(\begin{matrix} 2i \sin(t) & -2i \sin(t) \\ 0 & 0 \end{matrix} \right) + e^{2i \cos(t)}$$

$$e^{2i \sin(t)} = \cos(2 \sin(t)) + i \sin(2 \sin(t))$$

$$e^{-2i \sin(t)} = \cos(2 \sin(t)) - i \sin(2 \sin(t))$$

$$= 2 \cos(2 \sin(t))$$

Segunda tentativa

Série de Taylor de uma exponencial

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\rho(t) = e^{2\cos(t)}, \cos(2\sin(t))$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi kt/T}$$

$$X_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi kt/T} dt$$

$$\cos(2\sin(t)) = \frac{e^{i \cdot (2\sin(t))} + e^{-i(2\sin(t))}}{2}$$

$$\rho(t) = \frac{e^{2\cos(t) + 2i\sin(t)} + e^{2\cos(t) - 2i\sin(t)}}{2}$$

$$X_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi kt/T} dt$$

$$T > 2\pi$$

$$X_k = \frac{1}{2\pi} \cdot \int_0^{2\pi} e^{2\cos(t) + 2i\sin(t) - ikt} dt + \int_0^{2\pi} e^{2\cos(t) + 2i\sin(t) - ikt} dt$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sum_{n=0}^{\infty} \frac{(2\cos(t) + 2i\sin(t) - ikt)^n}{n!} = \sum_{n=0}^{\infty} \frac{(2e^{it} - ikt)^n}{n!}$$

$$\cos(t) + i\sin(t) = e^{it}$$

$$= \sum_{n=0}^{\infty} \frac{(2e^{it} - ikt)^n}{n!}$$

$$\sum_{n=0}^{\infty} \int_0^{2\pi} \frac{(2e^{it} - ikt)^n}{n!} + \frac{(2e^{-it} - ikt)^n}{n!} dt$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_0^{2\pi} (2e^{it} - ikt)^n + (2e^{-it} - ikt)^n dt$$

↓

L

$$L. \left\{ \int_0^{2\pi} (2e^{it} - ikt)^n dt + \int_0^{2\pi} (2e^{-it} - ikt)^n dt \right.$$

terceira Tentativa

$$f(t) = e^{2\cos(t)} \cdot \cos(2\sin(t))$$

$$\cos(2\sin(t)) = \frac{e^{i2\sin(t)} + e^{-i2\sin(t)}}{2}$$

$$f(t) = e^{2\cos(t)} \cdot \left(\frac{e^{i2\sin(t)} + e^{-i2\sin(t)}}{2} \right)$$

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

$$f(t) = e^{(e^{it} + e^{-it})} \cdot \left(\frac{(e^{it} - e^{-it})(e^{-it} - e^{it})}{2} \right)$$

$$f(t) = \frac{(e^{2e^{it}} + e^{2e^{-it}})}{2}$$

Série de Taylor de uma exponencial

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$p(t) = e^{2e^{it}}$$

$$q(t) = e^{2e^{-it}}$$

$$f(t) = \frac{p(t) + q(t)}{2}$$

$$p(t) = e^x, \quad x = 2e^{it}; \quad q(t) = e^y, \quad y = 2e^{-it}$$

$$p(t) = \sum_{n=0}^{\infty} \frac{(2e^{it})^n}{n!}$$

$$q(t) = \sum_{n=0}^{\infty} \frac{(2e^{-it})^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(2e^{it})^n}{n!}$$

=

$$f(t) = \frac{1}{2} \cdot \left\{ \sum_{n=0}^{\infty} \frac{(2e^{it})^n}{n!} + \sum_{n=0}^{\infty} \frac{(2e^{-it})^n}{n!} \right\}$$

$$f(t) = \frac{1}{2} \cdot \left\{ \sum_{n=0}^{\infty} \frac{(2e^{it})^n + (2e^{-it})^n}{n!} \right\}$$

$$n=0$$

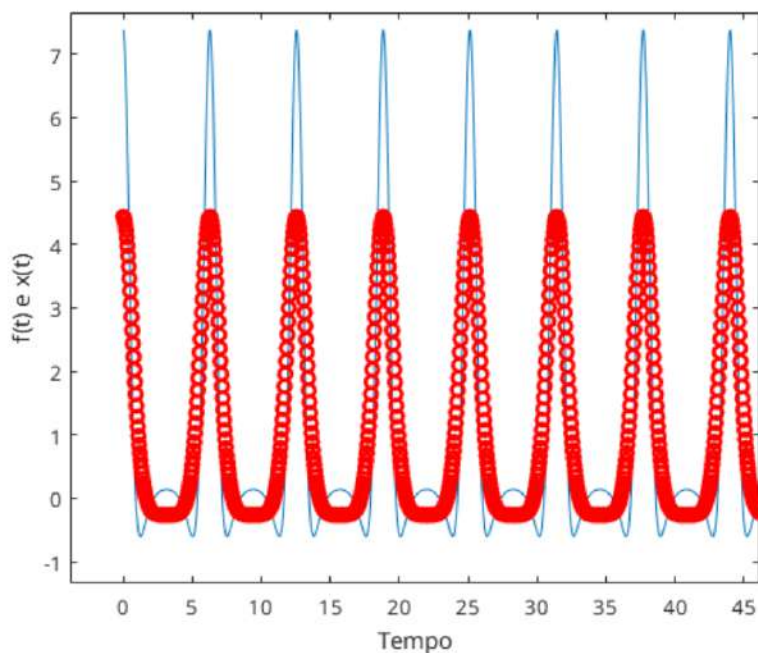
$$\frac{1}{d} \cdot \left\{ \frac{1+1}{1} \right\} = 1$$

$$f(t) = 1 + \frac{1}{d} \cdot \sum_{n=1}^{\infty} \frac{d e^{i t n} + d e^{-i t n}}{n!}$$

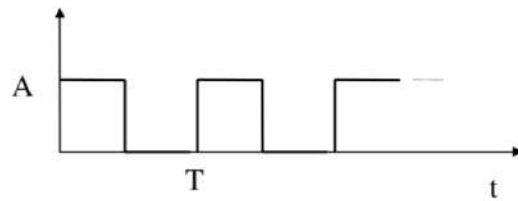
$$f(t) = 1 + \sum_{n=1}^{\infty} \frac{e^{i t n} + e^{-i t n}}{n!}$$

$$\cos(n t) = \frac{e^{i t n} + e^{-i t n}}{2}$$

$$f(t) = 1 + \sum_{n=1}^{\infty} \frac{2 \cdot \cos(n t)}{n!}$$



1.4 Calcular os coeficientes da série de Fourier complexa para uma onda quadrada de período T e amplitude pico-a-pico A mostrada na figura abaixo:



$$x(t) = \sum_{k=-\infty}^{\infty} x_k \cdot e^{i2\pi k t / T}$$

$$x_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi k t / T} dt$$

$$x(t) = \begin{cases} A, & T \cdot p \leq t \leq T \cdot p + \frac{T}{2} \\ 0, & T \cdot p + \frac{T}{2} < t < T \cdot (p+1) \end{cases} \quad p \in \mathbb{N}$$

com $p=0$

$$x(t) = \begin{cases} A, & 0 \leq t \leq T/2 \\ 0, & T/2 < t < T \end{cases}$$

$$x_k = \frac{1}{T} \int_0^{T/2} A \cdot e^{-i2\pi k t / T} dt = \frac{A \cdot T}{T} \left[\frac{e^{-i2\pi k t / T}}{-i2\pi k} \right]_0^{T/2}$$

$$= \frac{A}{i2\pi k} \left[e^{-i2\pi k T/2T} - 1 \right] = \frac{A}{i2\pi k} \cdot \left[e^{-i\pi k} - 1 \right]$$

$$e^{-i\pi k} = \cos(\pi k) - i \sin(\pi k)$$

-1 para k ímpar

1 para k par

$$x_k = \begin{cases} \frac{A}{i\pi k}, & k \text{ ímpar} \\ 0, & k \text{ par} \end{cases} \quad - \frac{Ai}{\pi k}, \quad |x_k| = \frac{|A|}{\pi \cdot |k|}$$

$$x(t) = \sum_{k=1,3,5,\dots}^{\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T} + \sum_{k=-1,-3,-5,\dots}^{-\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T}$$

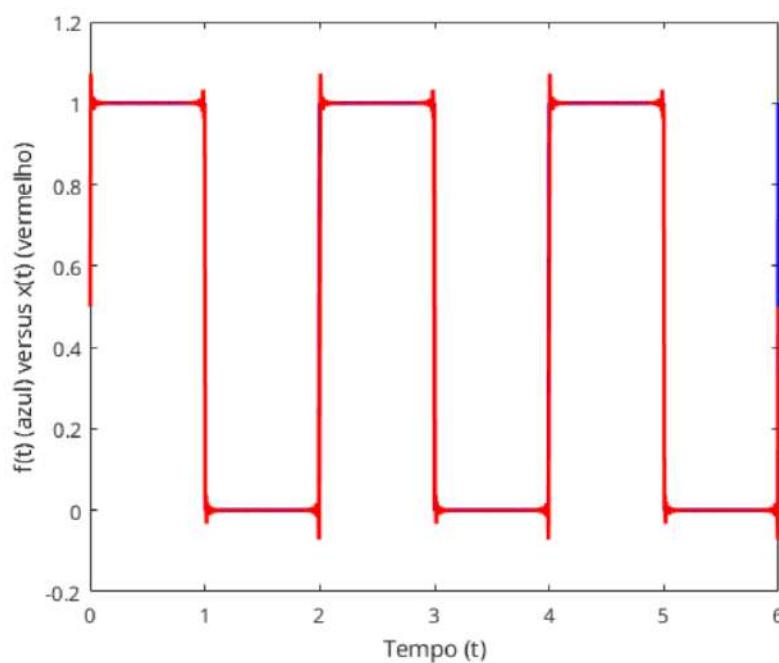
$$x(t) = \sum_{k=1,3,5,\dots}^{\infty} \frac{A}{i\pi k} \cdot e^{i2\pi k t/T} - \frac{A}{i\pi k} \cdot e^{-i2\pi k t/T}$$

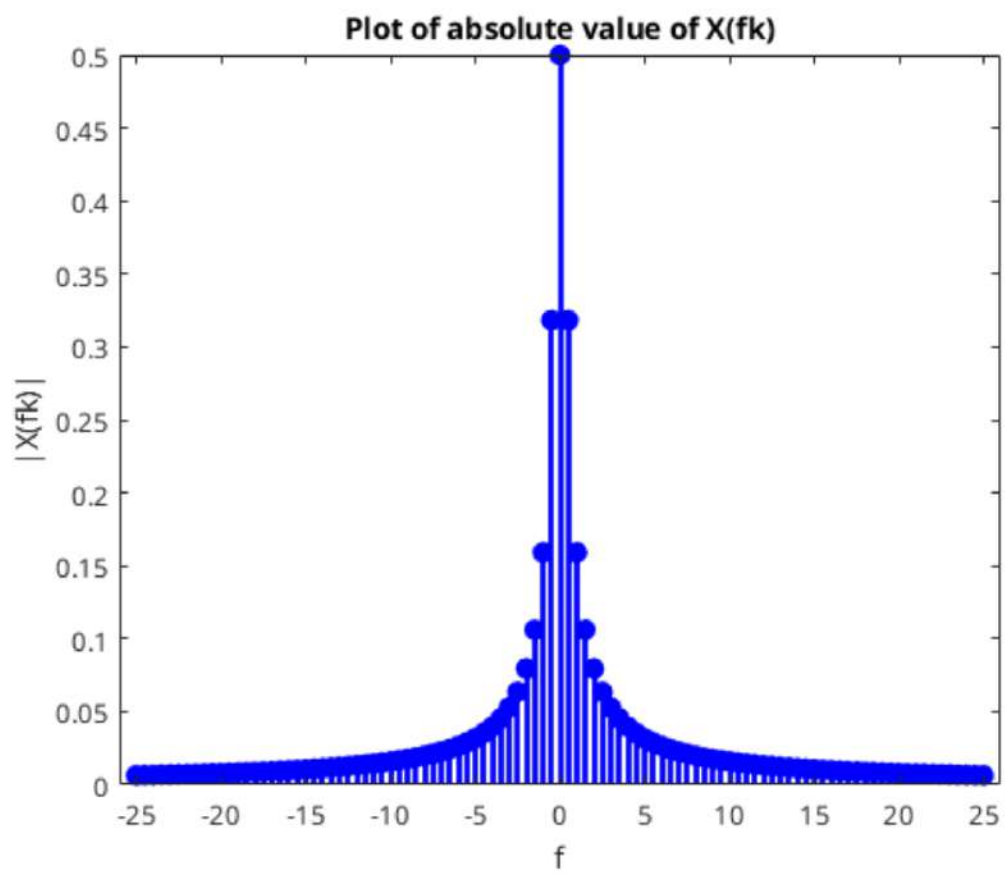
$$x(t) = \frac{A}{i\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{1}{k} \cdot (e^{i2\pi k t/T} - e^{-i2\pi k t/T})$$

$$x(t) = \frac{A}{\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{2}{k} \cdot (\sin(2\pi k t/T))$$

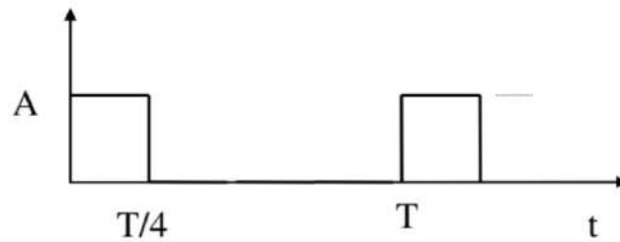
$$x_0 = \frac{1}{T} \cdot \int_0^{T/2} A \cdot e^{-i2\pi \cdot 0 \cdot t/T} = \frac{A \cdot T}{T \cdot 2} = \frac{A}{2}$$

$$x(t) = \frac{A}{2} + \frac{A}{\pi} \cdot \sum_{k=1,3,5,\dots}^{\infty} \frac{2}{k} \cdot (\sin(2\pi k t/T))$$





1.5 Calcular os coeficientes da série de Fourier complexa para a onda de período T e amplitude pico-a-pico A mostrada na figura abaixo:



$$x(t) = \begin{cases} A, & T \cdot p \leq t \leq T \cdot p + T/4 \\ 0, & T \cdot p + T/4 < t < T \cdot (p+1) \end{cases} \quad p \in \mathbb{N}$$

$$p=0$$

$$x(t) = \begin{cases} A, & 0 \leq t \leq T/4 \\ 0, & T/4 < t < T \end{cases}$$

$$x_k = \frac{1}{T} \int_0^{T/4} A \cdot e^{-i2\pi kt/T} dt = \frac{A}{T} \cdot \left[\frac{e^{-i2\pi kt/T}}{-i2\pi k/T} \right]_0^{T/4}$$

$$\frac{A}{T} \cdot \frac{T}{-i2\pi k} \cdot [e^{-i2\pi k T/4} - 1] = \frac{-A}{i2\pi k} \cdot [e^{-i\pi k/2} - 1]$$

$$e^{-i\pi k/2} = \cos(\pi k/2) - i \sin(\pi k/2)$$

$$k=0 \quad b=1$$

$$k=1 \quad = -i$$

$$k=2 \quad = -1$$

$$k=3 \quad = i$$

$$x_k = \begin{cases} \frac{A}{4}, & k=0 \\ 0, & k \% 4 = 0, k \neq 0 \\ \frac{A}{i2\pi k} (i+1), & k=1 \\ \frac{A}{i\pi k}, & k=2 \\ \frac{A}{i2\pi k} (1-i), & k=3 \end{cases}$$

$$x_0 = \frac{A}{T} \cdot \int_0^{T/4} 1 dt = \frac{A}{T} \cdot \frac{T}{4} = \frac{A}{4}$$

0, 1, 2, 3,
4, 5, 6, 7
8, 9, 10, 11

Intervalos se repetem a cada 4.

$$x_k = \begin{cases} \frac{A}{4}, & k=0 \\ 0, & 4k, k \neq 0 \\ \frac{A(i+1)}{i2\pi k}, & 4k+1 \\ \frac{A}{i2\pi k}, & 4k+2 \\ \frac{A(1-i)}{i2\pi k}, & 4k+3 \end{cases} \quad k = 0, 1, 2, 3, \dots$$

$$|x_k| = \frac{|A(i+1)|}{|i2\pi k|} = \frac{|A| \cdot \sqrt{1^2+1^2}}{|i| \cdot |2\pi k|} = \frac{|A| \cdot \sqrt{2}}{1 \cdot 2\pi |k|}$$

$\sqrt{1^2+1^2} = \sqrt{2}$
 $\begin{matrix} \nearrow & \nwarrow \\ 1 & 1 \\ \downarrow & \downarrow \\ i & 1 \end{matrix}$

$$\frac{|A| \cdot \sqrt{2}}{2\pi |k|}$$

$$k \% 4 = 2$$

$$|x_k| = \frac{|A|}{|i2\pi k|} = \frac{|A|}{2\pi \cdot |k|}$$

$$k \% 4 = 3$$

$$\sqrt{1+1^2} = \sqrt{2}$$

$$|x_k| = \frac{|A| \cdot |i-1|}{|i| 2\pi |k|} = \frac{|A| \cdot \sqrt{2}}{2\pi \cdot |k|}$$

$$|x_k| = \begin{cases} 0, & k \% 4 = 0 \\ \frac{|A| \cdot \sqrt{2}}{2\pi \cdot |k|}, & k \% 4 = 1 \vee k \% 4 = 3 \\ \frac{|A|}{2\pi |k|}, & k \% 4 = 2 \end{cases}$$

$$\frac{-A}{i2\pi k} \cdot [e^{-i\pi k/2} - 1] = \frac{A}{i2\pi k} \cdot [1 - e^{-i\pi k/2}]$$

$$e^{-i\pi k/2} = \cos(\pi k/2) - i \sin(\pi k/2)$$

$k \text{ par}$	$k \% 4 = 0$	1	$x_k = 0$
	$k \% 4 = 1$	-i	$x_k = A / (i2\pi k) \cdot (1+i)$
	$k \% 4 = 2$	-1	$x_k = A / i2\pi k$
	$k \% 4 = 3$	i	$x_k = (A / i2\pi k) \cdot (1-i)$

$$\frac{A \cdot e^{-i\pi k/4}}{\pi k} \cdot \left(\frac{e^{i\pi k/4} - e^{-i\pi k/4}}{2i} \right)$$

$$\frac{A \cdot e^{-i\pi k/4}}{\pi k} \cdot (\sin(\pi k/4))$$

For trigonometric

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left[a_k \cos\left(k \frac{2\pi}{T} t\right) + b_k \sin\left(k \frac{2\pi}{T} t\right) \right]$$

$$\frac{A}{i2\pi k} \left(1 - e^{-i\pi k/2} \right) = \lambda_k$$

$$a_k = \lambda_k + \lambda_{-k} \quad b_k = j(\lambda_k - \lambda_{-k})$$

$$a_k = \frac{A}{i2\pi k} \left[1 - e^{-i\pi k/2} \right] + \frac{A}{-i2\pi k} \left[1 - e^{i\pi k/2} \right]$$

$$a_k = \frac{A}{i2\pi k} \left[1 - e^{-i\pi k/2} - 1 + e^{i\pi k/2} \right]$$

$$a_k = \frac{A}{i2\pi k} \left[e^{i\pi k/2} - e^{-i\pi k/2} \right]$$

$$a_k = \frac{A}{\pi k} \sin(\pi k/2)$$

$$b_k = \frac{Ai}{i2\pi k} \left[1 - e^{-i\pi k/2} + 1 - e^{i\pi k/2} \right]$$

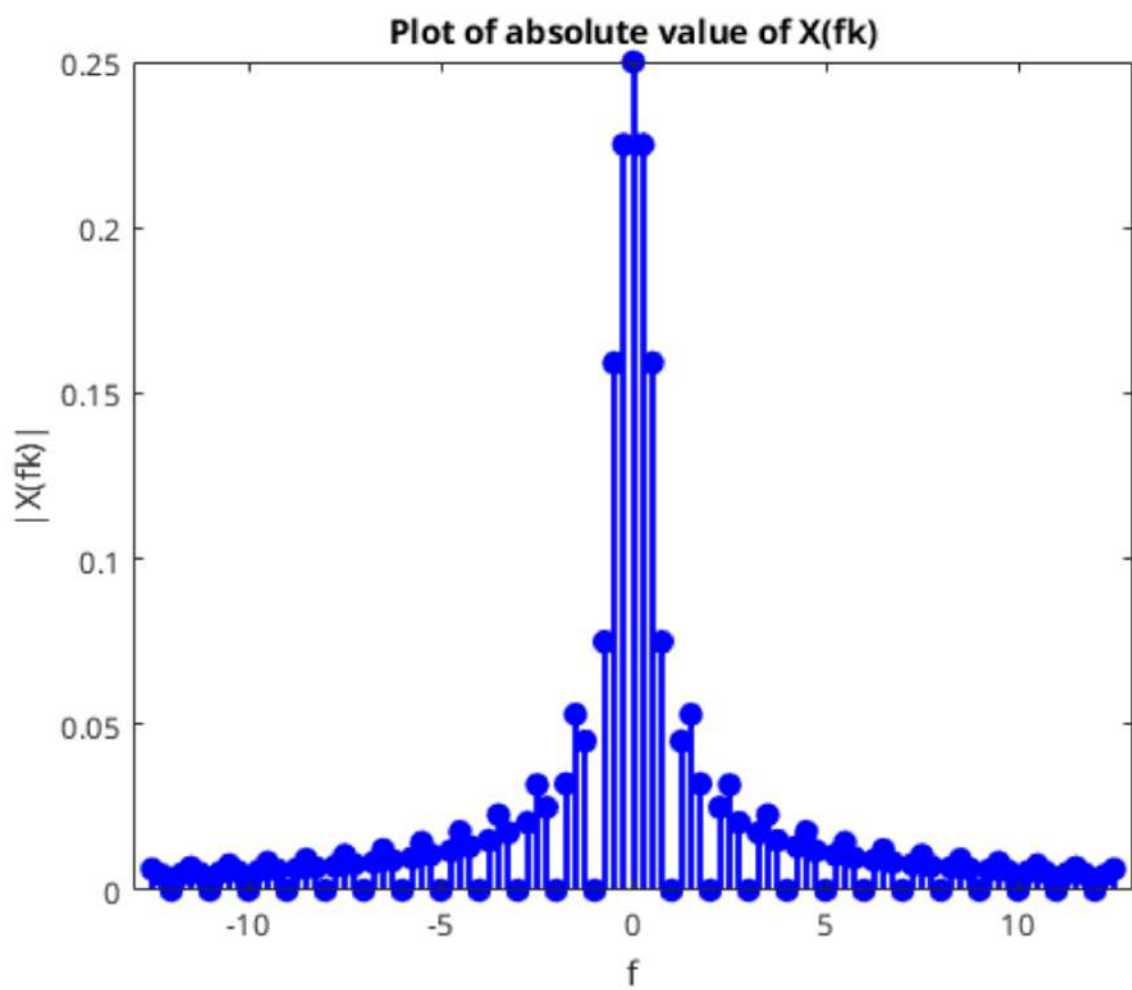
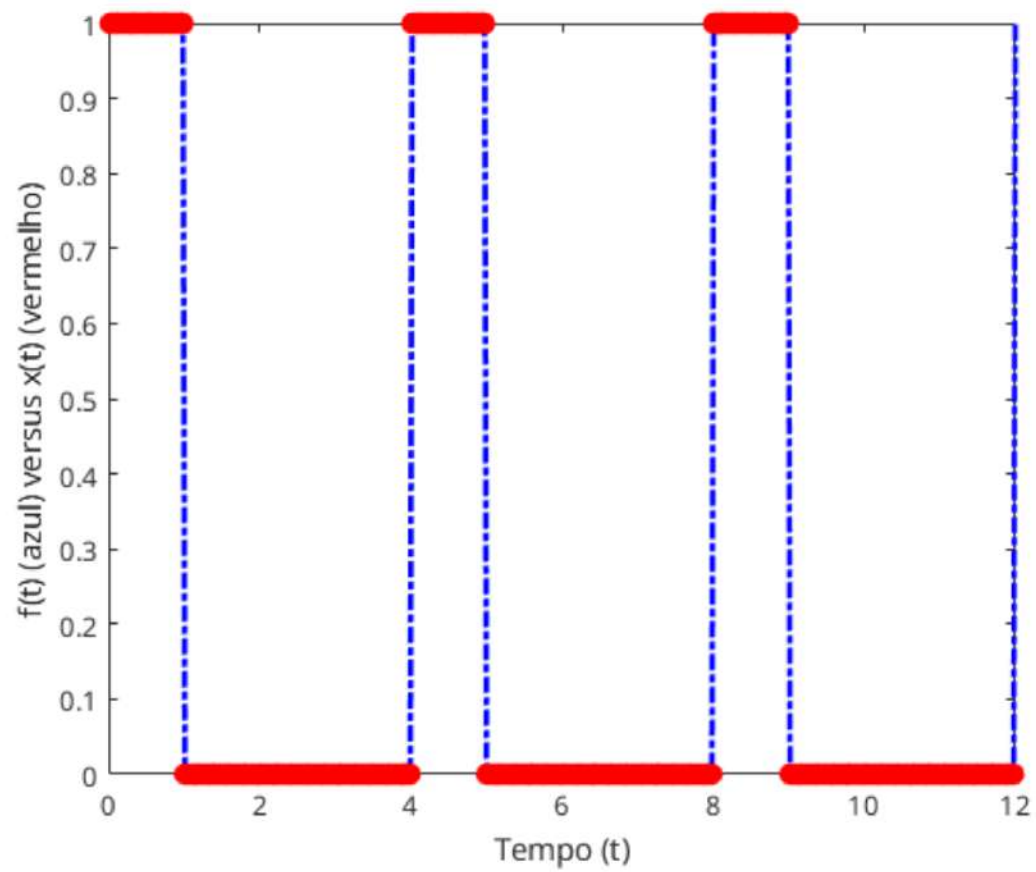
$$b_k = \frac{A}{\pi k} + \frac{-A}{2\pi k} [e^{i\pi k/2} + e^{-i\pi k/2}]$$

$$b_k = \frac{A}{\pi k} = \frac{A}{\pi k} \cdot \cos(\pi k/2)$$

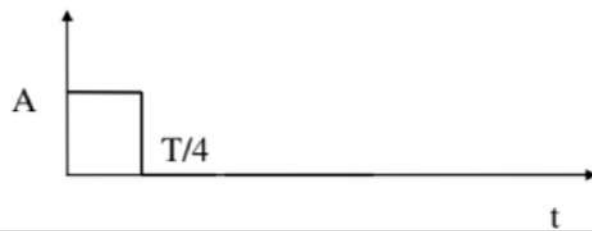
$$b_k = \frac{A}{\pi k} (1 - \cos(\pi k/2))$$

$$x(t) = \frac{x_0}{2} + \sum_{k=1}^{\infty} [a_k \cos(k \frac{2\pi}{T} t) + b_k \sin(k \frac{2\pi}{T} t)]$$

$$x(t) = \frac{A}{4} + \sum_{k=1}^{\infty} \frac{A}{\pi k} \cdot \left[\sin\left(\frac{\pi k}{2}\right) \cdot \cos\left(\frac{k 2\pi}{T} t\right) + (1 - \cos(\pi k/2)) \cdot \sin\left(k \frac{2\pi}{T} t\right) \right]$$



1.6 Calcular a transformada (integral) de Fourier do pulso retangular abaixo. Se possível, traçar os resultados dos itens anteriores usando o programa MATLAB (traçar somente os valores absolutos).



Correção do item 1.6 13/06/2025

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{i2\pi ft} df$$

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i2\pi ft} dt$$

$$x(t) = \begin{cases} A, & 0 \leq t \leq T/4 \\ 0, & t < 0 \vee t > T/4 \end{cases}$$

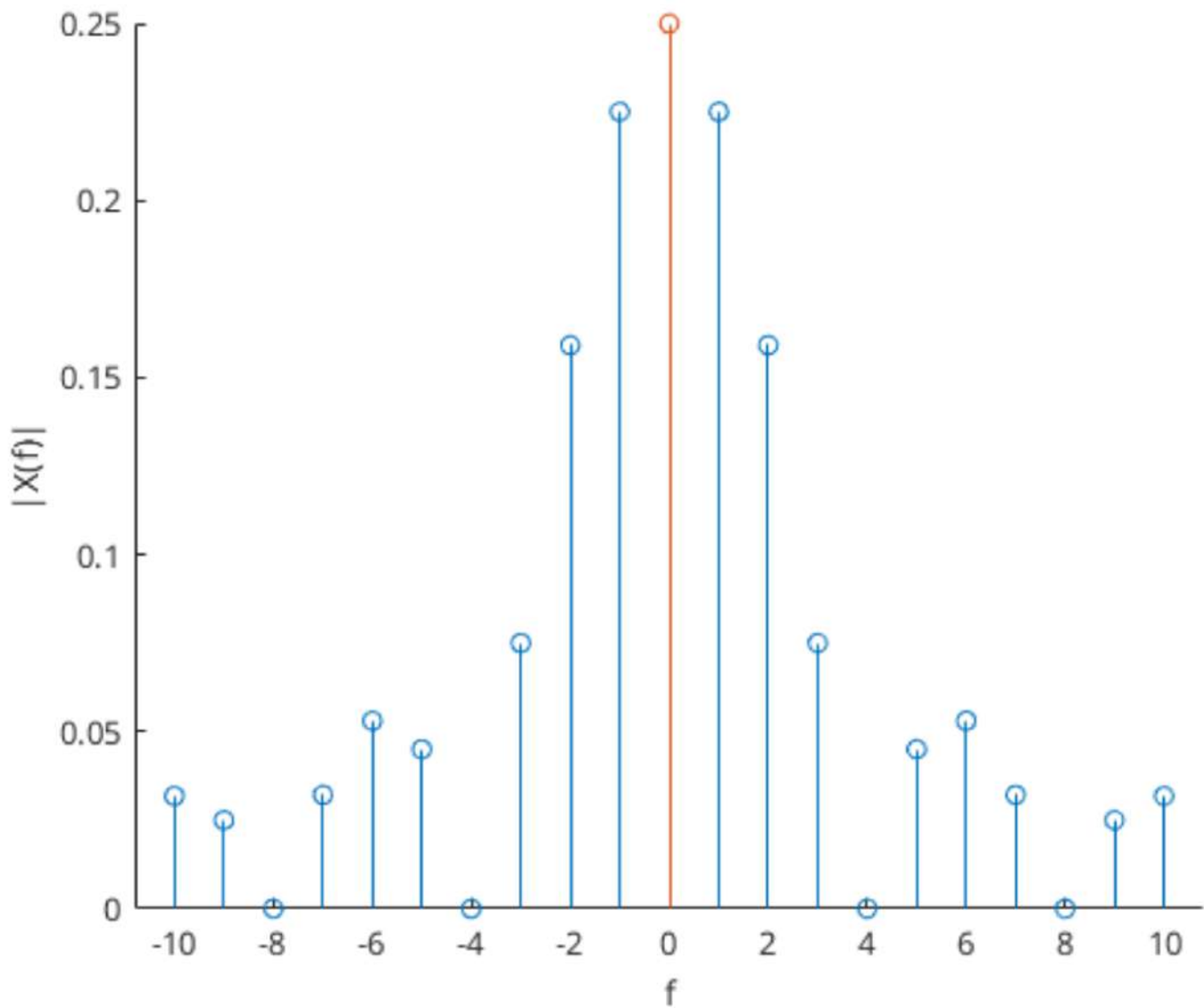
$$X(f) = \int_0^{T/4} A \cdot e^{-i2\pi ft} dt = A \cdot \left[\frac{e^{-i2\pi ft}}{-i2\pi f} \right]_0^{T/4}$$

$$= \frac{A}{-i2\pi f} \cdot [e^{-i2\pi f T/4} - 1]$$

$$= \frac{A}{-i2\pi f} \cdot [e^{-i\pi f T/2} - 1] = \frac{Ai}{2\pi f} \cdot [e^{i\pi f T/2} - 1]$$

$$|X(0)| = \int_0^{T/4} A \cdot e^{-i2\pi 0 t} dt = A \cdot \int_0^{T/4} dt$$

$$= A \cdot (t)_0^{T/4} = \frac{A \cdot T}{4}$$



Questão 1.7

1.7 Provar o teorema da convolução na forma:

$$x(t)h(t) \Leftrightarrow X(f) * H(f)$$

Transformada de Fourier

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{i2\pi ft} df$$

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i2\pi ft} dt$$

Conclusão

$$\begin{aligned} x(t) * h(t) &= \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau = h(t) * x(t) \end{aligned}$$

$$x(t) \cdot h(t) \xrightarrow{F} X(f) * H(f)$$

$$X(f) * H(f) = \int_{-\infty}^{\infty} X(\tau) \cdot H(f-\tau) d\tau$$

$$\int_{-\infty}^{\infty} x(t) \cdot h(t) \cdot e^{-i2\pi ft} dt = \int_{-\infty}^{\infty} X(\tau) \cdot H(f-\tau) d\tau$$

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi (f-\tau)t} dt \right] \cdot X(\tau) d\tau$$

$$\int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi ft} \cdot e^{i2\pi \tau t} dt \right] \cdot X(\tau) d\tau$$

Usando o teorema de Euler:

$$\int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi ft} \cdot \left[\int_{-\infty}^{\infty} X(\tau) \cdot e^{i2\pi \tau t} d\tau \right] dt$$

Relembrando

$$x(t) = \int_{-\infty}^{\infty} X(\tau) \cdot e^{i2\pi \tau t} d\tau$$

Logo

o sinal é $x(t)$

$$= \int_{-\infty}^{\infty} h(t) \cdot e^{-i2\pi ft} \cdot x(t) dt$$

$$= \int_{-\infty}^{\infty} x(t) \cdot h(t) \cdot e^{-i2\pi ft} \cdot dt$$

Provando que

$$x(t) \cdot h(t) \stackrel{F}{\Leftrightarrow} X(f) * H(f)$$

Questão 1.8

1.8 Mostrar que os coeficientes da Série de Fourier podem ser aproximados pela expressão:

$$X_k = \frac{1}{N} \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}$$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{i2\pi kt/T}$$

$$X_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot e^{-i2\pi kt/T} dt = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi kt/T} dt$$

A soma de Riemann ^{à esquerda} é definida como

$$\sum_{i=0}^{n-1} f(x_i) \cdot (x_{i+1} - x_i)$$

Seja $f(t)$ uma função definida no intervalo $[a, b]$.

A integral definida de Riemann é definida como o limite da soma de Riemann quando n tende ao infinito.

$$\lim_{n \rightarrow \infty} \sum_{i=0}^{n-1} f(x_i) \cdot (x_{i+1} - x_i) = \sum_{i=0}^{\infty} f(x_i) \cdot (x_{i+1} - x_i)$$

$\quad \quad \quad \checkmark \quad \quad \quad \checkmark$
 $\quad \quad \quad \Delta x \quad \quad \quad \Delta x$

$$= \int_a^b f(t) dt$$

Voltando a definição de X_k

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-j2\pi t/T} dt$$

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-j2\pi t/T} dt \quad f(t) = x(t) \cdot e^{-j2\pi t/T}$$

Reescrevendo a integral como uma soma de Riemann com n tendendo ao infinito

$$\sum_{i=0}^{\infty} f(x_i) \cdot \underbrace{(x_{i+1} - x_i)}_{\Delta x} = \sum_{n=0}^{\infty} f(t_n) \cdot \underbrace{(t_{n+1} - t_n)}_{\Delta t}$$

$$\Delta t = \frac{T-0}{N}$$

$$t_n = n \Delta t$$

$$x_n = x(t_n)$$

Número de intervalos N

$$\sum_{n=0}^{\infty} f(t_n) \cdot \Delta t = \lim_{N \rightarrow \infty} \sum_{n=0}^N f(t_n) \cdot \Delta t$$

$$\approx \sum_{n=0}^{N-1} f(t_n) \cdot \frac{T}{N} = \frac{T}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k t_n T / N}$$

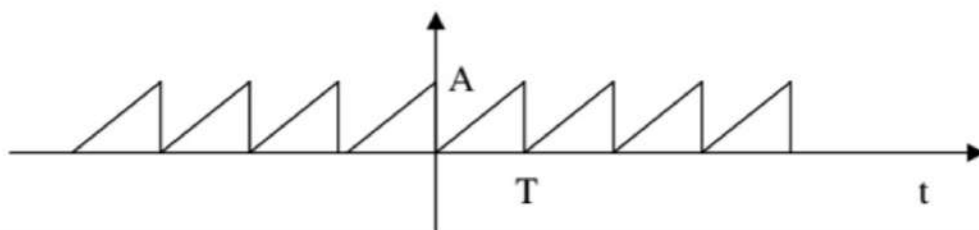
$$= \frac{T}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n T / N T} = \frac{T}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

$$X_k = \frac{1}{T} \int_0^T f(t) dt \approx \frac{1}{T} \cdot \frac{T}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

$$\approx \frac{1}{N} \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N}$$

Questão 1.9

1.9 Calcular a série de Fourier de um sinal do tipo dente de serra usando a aproximação acima e comparar com a Série de Fourier analítica. Comparar os espectros graficamente. Usar $A = 3$ e $T = 7$



$$\hat{x}_k = \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \cdot e^{-i2\pi k n / N} \quad , \quad x_n = x(t_n)$$

$$t_n = n \cdot \Delta t = n \cdot \frac{T}{N}$$

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-i2\pi k t / T} dt$$

$$x(t) = \frac{A t}{T} \quad x\left(t = n \frac{T}{N}\right) = A \cdot \frac{n T}{N \cdot T}$$

$$\hat{x}_k = \frac{A}{N^2} \cdot \sum_{n=0}^{N-1} n \cdot e^{-i2\pi k n / N} \quad \left. \vphantom{\sum_{n=0}^{N-1}} \right\} = \frac{A n}{N}$$

$$x(t) = \sum_{k=-\infty}^{\infty} \left[\frac{A}{N^2} \cdot \sum_{n=0}^{N-1} n \cdot e^{-j2\pi k n / N} \right] \cdot e^{j2\pi k t / T}$$

Après Fourier Analitico

$$X_k = \frac{1}{T} \cdot \int_0^T x(t) \cdot e^{-j2\pi k t / T} dt$$

$$X_k = \frac{1}{T} \cdot \int_0^T \frac{A t}{T} \cdot e^{-j2\pi k t / T} dt$$

$$X_k = \frac{A}{T^2} \cdot \int_0^T t \cdot e^{-j2\pi k t / T} dt$$

$$\int u dv = u \cdot v - \int v du$$

$$u = t \quad du = dt \quad dv = e^{-j2\pi k t / T} \quad v = \frac{T \cdot e^{-j2\pi k t / T}}{-j2\pi k}$$

$$\left[\frac{T \cdot t \cdot e^{-j2\pi k t / T}}{-j2\pi k} \right]_0^T - \int_0^T \frac{T \cdot e^{-j2\pi k t / T}}{-j2\pi k} dt$$

$$\frac{-T^2 \cdot e^{-j2\pi k}}{j2\pi k} + \frac{T}{j2\pi k} \cdot \left[e^{-j2\pi k t / T} \right]_0^T$$

$$\frac{T}{i2\pi k} \left(-T e^{-i2\pi k} + e^{-i2\pi k} - 1 \right)$$

$$e^{-i2\pi k} = \cos(2\pi k) - i \sin(2\pi k), k=0,1,2,\dots$$

$$\frac{T}{i2\pi k} \begin{pmatrix} -T & 1 & -1 \end{pmatrix} = \frac{-T^2}{i2\pi k}$$

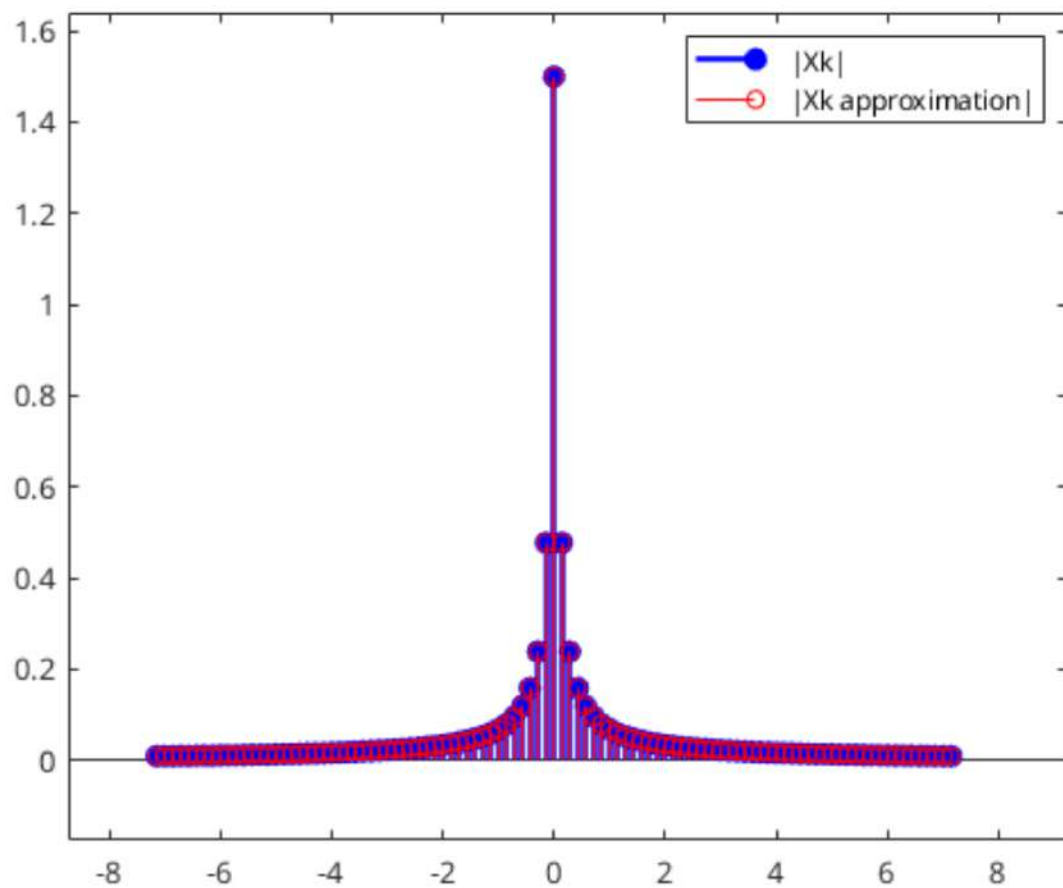
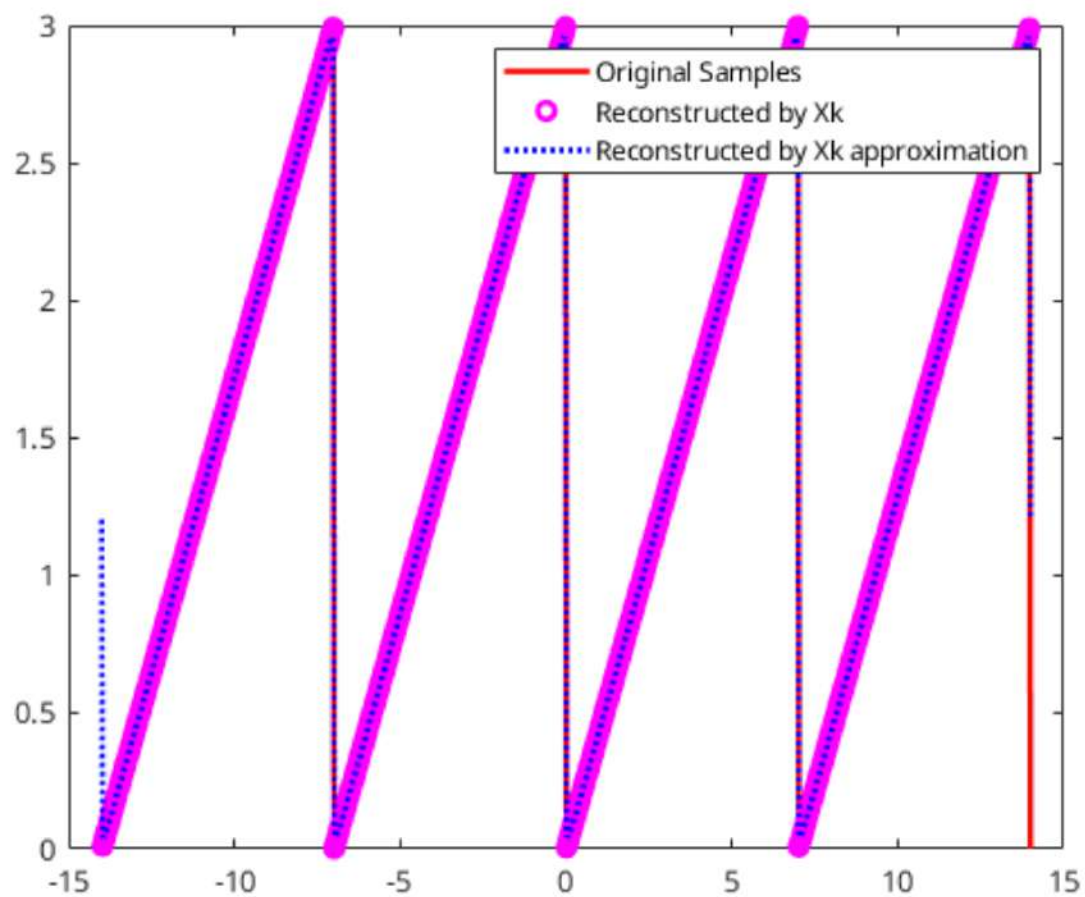
Voltando a equação original

$$X_k = \frac{A}{T^2} \cdot \frac{-T^2}{i2\pi k} = \frac{A i}{2\pi k}$$

$$X_0 = \frac{1}{T} \int_0^T \frac{A}{T} dt = \frac{A}{T^2} \cdot \left[\frac{t^2}{2} \right]_0^T$$

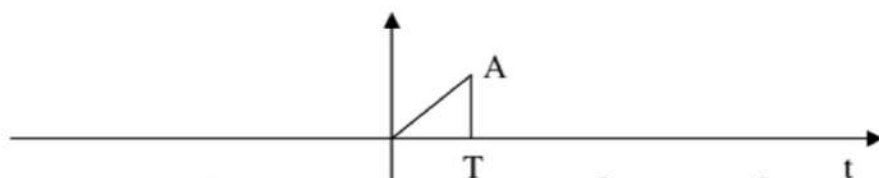
$$= \frac{A}{2}$$

$$x(t) = \frac{A}{2} + \sum_{k=1}^{\infty} \frac{A i}{2\pi k} e^{j2\pi k t / T} + \sum_{k=-\infty}^{-1} \frac{A i}{2\pi k} e^{j2\pi k t / T}$$



Questão 10

1.10 Calcular a Transformada de Fourier de um pulso triangular (um período do sinal do item anterior) usando a aproximação da SF e a fórmula de Poisson.



$$\hat{x}_k = \frac{1}{N} \sum_{n=0}^{N-1} x_n e^{-i 2 \pi k n / N}$$

$$t_n = n \cdot \Delta t$$

$$= n \cdot \frac{T}{N}$$

$$x(t) = \frac{A \cdot t}{T}$$

$$x(t) \Big|_{t=t_n} = \frac{A \cdot n \cdot T}{T \cdot N}$$

$$= \frac{A \cdot n}{N}$$

Fórmula de Poisson:

$$\sum_{n=-\infty}^{\infty} x(t + nT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} x(nT) \cdot e^{i 2 \pi k n T}$$

1- Calcular $\hat{X}_k$

$$\hat{X}_k = \frac{1}{N} \cdot \sum_{m=0}^{N-1} x_m \cdot e^{-i2\pi k m / N}$$

$$\hat{X}_k = \frac{1}{N} \cdot \sum_{m=0}^{N-1} \frac{A \cdot m}{N} \cdot e^{-i2\pi k m / N}$$

$$= \frac{A}{N^2} \cdot \sum_{m=0}^{N-1} m \cdot e^{-i2\pi k m / N} = \frac{A}{N^2} \cdot \sum_{m=1}^{N-1} m \cdot e^{-i2\pi k m / N}$$

2- Poisson

A transformada de Fourier de um sinal periódico pode ser dada por:

$$x(t) = \sum_{k=-\infty}^{\infty} x\left(\frac{k}{T}\right) \cdot \delta\left(t - \frac{k}{T}\right)$$

Os coeficientes $\hat{X}_k$ correspondem a amostras da transformada de Fourier $X(f)$ nos pontos $\frac{k}{T}$

3-

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-i2\pi f t} dt$$

Fórmula de Poisson:

$$\sum_{n=-\infty}^{\infty} x(t + nT) = \frac{1}{T} \cdot \sum_{n=-\infty}^{\infty} X(nf_0) \cdot e^{i2\pi f_0 n t}$$

$$X_n = \frac{A}{N^2} \sum_{n=1}^{N-1} e^{-i2\pi k n / N}$$

$$X(f) = \int_{-\infty}^{\infty} \frac{1}{T} \cdot \frac{A}{N^2} \cdot \sum_{n=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} n \cdot e^{-i2\pi k n / N} \cdot e^{i2\pi f t} dt$$

com $T \rightarrow \infty$ e $f_0 = \frac{1}{T}$

Questão 11

1.11 Demonstre que a transformada de Fourier de: $a + b\delta(t-t_0) + c\delta'(t-t_1)$ é $2\pi a\delta(\omega) + be^{-i\omega t_0} + ic\omega e^{-i\omega t_1}$

$$a + b\delta(t-t_0) + c\delta'(t-t_1)$$

$$\xrightarrow{F} 2\pi a\delta(\omega) + be^{-i\omega t_0} + ic\omega e^{-i\omega t_1}$$

$$= F(a) + F(b\delta(t-t_0)) + F(c\delta'(t-t_1))$$

$$= \underbrace{a \cdot F(1)}_{(1)} + b \cdot \underbrace{F(\delta(t-t_0))}_{(2)} + ic \cdot \underbrace{F(\delta'(t-t_1))}_{(3)}$$

$$(1) \delta(t) = 2\pi\delta(\omega)$$

$$(2) F[\delta(t-t_0)] = \int_{-\infty}^{\infty} \delta(t-t_0) \cdot e^{-i2\pi ft} dt$$

$$= e^{-i2\pi ft_0} = e^{-i\omega t_0}$$

$$\textcircled{3} \mathcal{F}[\delta'(t-t_1)]$$

Usando a propriedade da diferenciação

$$\mathcal{F}[\dot{x}(t)] = j2\pi f \cdot X(f)$$

$$\mathcal{F}[\dot{\delta}(t-t_1)] = j2\pi f \cdot \mathcal{F}[\delta(t-t_1)]$$

$$= j2\pi f \cdot e^{j2\pi f t_1} = j\omega \cdot e^{j\omega t_1}$$

Juntando ①, ② e ③ na equação original

$$a \cdot \textcircled{1} + b \cdot \textcircled{2} + ic \cdot \textcircled{3}$$

$$= a \cdot 2\pi \cdot \delta(\omega) + b \cdot e^{j\omega t_0} + ic \cdot j\omega \cdot e^{j\omega t_1}$$

12

$$\ddot{x}(t) + 5\dot{x}(t) + 6x(t) = \delta(t) \quad x(t) = (e^{-2t} + e^{-3t})u(t)$$

$$F\{x(t)\} = X(f)$$

$$F\{\dot{x}(t)\} = j2\pi f X(f)$$

$$F\{\ddot{x}(t)\} = (j2\pi f)^2 \cdot X(f) = -(2\pi f)^2 \cdot X(f)$$

$$F\{\delta(t)\} = 1$$

$$-(2\pi f)^2 \cdot X(f) + 10j\pi f X(f) + 6X(f) = 1$$

$$X(f) (-2\pi^2 f^2 + 10j\pi f + 6) = 1$$

$$X(f) = \frac{1}{-2\pi^2 f^2 + 10j\pi f + 6} = \frac{1}{j} \cdot \frac{1}{-2\pi^2 f^2 + 5j\pi f + 3}$$

Achando os polos de $X(f)$

$$2\pi^2 f^2 - 5j\pi f - 3 = 0$$

$$\frac{-(-5j\pi) \pm \sqrt{(-5j\pi)^2 - 4 \cdot 2\pi^2 \cdot (-3)}}{4\pi^2}$$

$$\frac{5j\pi \pm \sqrt{-25\pi^2 + 24\pi^2}}{4\pi^2} = \frac{5j\pi \pm (j\pi)}{4\pi^2}$$

$$= \frac{5j \pm j}{4\pi} \quad \begin{cases} p_1 = \frac{j}{\pi} \\ p_2 = \frac{3j}{2\pi} \end{cases}$$

$$\left(l - \frac{j}{\pi}\right) \cdot \left(l - \frac{3j}{2\pi}\right) = \left(\frac{l\pi - j}{\pi}\right) \cdot \left(\frac{l2\pi - 3j}{2\pi}\right)$$

$$\frac{l^2 2\pi^2 - 3j\pi l - 2j\pi l + 3}{2\pi^2} = \frac{2\pi^2 l^2 - 5j\pi l + 3}{2\pi^2}$$

Voltando a $X(l)$

$$X(l) = \frac{1}{\left(l - \frac{j}{\pi}\right) \cdot \left(l - \frac{3j}{2\pi}\right)} \cdot \frac{1}{2 \cdot 2\pi^2}$$

$$X(l) = \frac{1}{4\pi^2} \cdot \frac{1}{\left(l - \frac{j}{\pi}\right) \cdot \left(l - \frac{3j}{2\pi}\right)} = \frac{1}{4\pi^2} \cdot \frac{1}{(l-d) \cdot \left(l - \frac{3d}{2}\right)}$$

$$\frac{j}{\pi} = d$$

Pelo princípio da linearidade

$$p \cdot x(t) \stackrel{F}{=} 0 \quad p \cdot X(l)$$

$$p = \frac{1}{4\pi^2}$$

$$X(l) = \frac{1}{(l-d) \cdot \left(l - \frac{3}{2}d\right)}$$

Por hipóteses parciais

$$X(f) = \frac{1}{(f-d)(f-\frac{3}{d}d)} = \frac{a}{f-d} + \frac{b}{f-\frac{3}{d}d}$$

$$a(f-\frac{3}{d}d) + b(f-d) = 1$$

$$f(a+b) + d(-\frac{3}{d}a - b) = 1$$

$$f(a+b) = 0 \quad a+b=0 \quad a=-b$$

$$d(-\frac{3}{d}a - b) = 1$$

$$-3a - db = \frac{2}{d}$$

$$-3(-b) - db = \frac{2}{d}$$

$$b = \frac{2}{d}$$

$$d = \frac{2}{b}$$

$$b = \frac{2\pi}{j}$$

$$a = -\frac{2\pi}{j}$$

$$= \frac{1}{(1-d)(1-\frac{3}{d}d)} = \frac{1}{1-d} + \frac{1}{1-\frac{3}{d}d}$$

$$= \frac{-2\pi}{(1-\frac{j}{n})} \cdot j + \frac{2\pi}{(1-\frac{3j}{2n})} \cdot j$$

$$= 2\pi \left(\frac{-1}{(1-\frac{j}{n})} + \frac{1}{(1-\frac{3j}{2n})} \right)$$

$$= 2\pi \left(\frac{-j}{(1-\frac{j}{n})} + \frac{j}{(1-\frac{3j}{2n})} \right)$$

$$= 2\pi j \left(\frac{-1}{(1-\frac{j}{n})} + \frac{1}{(1-\frac{3j}{2n})} \right)$$

$$= 2\pi j \left(\frac{1}{(1-\frac{j}{n})} - \frac{1}{(1-\frac{3j}{2n})} \right)$$

Voltando a P

$$P \cdot x(f) = \frac{1}{4\pi^2} \cdot 2\pi j \left(\frac{1}{(f-d)} - \frac{1}{(f-\frac{3}{2}d)} \right)$$

$$= \frac{j}{2\pi} \cdot \left(\frac{1}{(f-d)} - \frac{1}{(f-\frac{3}{2}d)} \right)$$

$$P = \frac{j}{2\pi} \quad x(f) = \frac{1}{(f-d)} - \frac{1}{(f-\frac{3}{2}d)}$$

$$x(t) = P \cdot \left\{ \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-d)} \cdot e^{j2\pi ft} df}_{(1)} - \underbrace{\int_{-\infty}^{\infty} \frac{1}{(f-\frac{3}{2}d)} \cdot e^{j2\pi ft} df}_{(2)} \right\}$$

$$(1) \int_{-\infty}^{\infty} \frac{e^{j2\pi ft}}{(f-d)} df = \int_{-\infty}^{\infty} \frac{e^{j2\pi ft}}{(f-\frac{j}{\pi})} df$$

Aplicando o Teorema dos resíduos

1- Identificação do polo

$$f = \frac{j}{s}$$

2- Fechamento de contorno

$t > 0$: fechamos o contorno no semiplano superior

$t < 0$: Como o polo $f = \frac{j}{s}$ se encontra apenas no semiplano superior, a integral real 0 e fechamos o semiplano inferior.

3- Cálculo do Resíduo

$$\text{Res}(f = j/\pi) = \lim_{s \rightarrow \frac{j}{\pi}} (s - j/\pi) \cdot \frac{e^{j2\pi ft}}{(s - j/\pi)}$$

$$= e^{j2\pi \frac{j}{\pi} t} = e^{-2t} \cdot u(t)$$

$u(t)$ adicionado, pois só é válido para $t \geq 0$.

4- Resultado

$$\int_{-\infty}^{\infty} \frac{e^{j2\pi ft}}{f - \frac{j}{\pi}} = 2\pi j \cdot \text{Res} \left(f = j/\pi \right)$$

$$= 2\pi j \cdot \left(e^{-2t} \cdot u(t) \right)$$

$$\begin{aligned} \textcircled{2} \int_{-\infty}^{\infty} \frac{e^{j2\pi ft}}{f - \frac{3j}{2\pi}} &= 2\pi j \cdot \text{Res} \left(f = 3j/2\pi \right) \\ &= 2\pi j \cdot e^{3t} \cdot u(t) \end{aligned}$$

Cálculo do Resíduo

$$\text{Res} \left(f = 3j/2\pi \right) = \lim_{f \rightarrow 3j/2\pi} \left(f - \frac{3j}{2\pi} \right) \cdot \frac{e^{j2\pi ft}}{\left(f - \frac{3j}{2\pi} \right)}$$

$$= e^{j2\pi \frac{3}{2\pi} t} = e^{-3t} \cdot u(t)$$

$u(t)$ adicionado, pois só é válido para $t \geq 0$.

Voltando ao cálculo de $x(t)$

$$P = \frac{j}{2\pi}$$

$$x(t) = P \cdot \left\{ \underbrace{\int_{-\infty}^{\infty} \frac{1}{(t-d)} \cdot e^{j2\pi ft} df}_{(1)} - \underbrace{\int_{-\infty}^{\infty} \frac{1}{(t-\frac{3}{2})} \cdot e^{j2\pi ft} df}_{(2)} \right\}$$

$$x(t) = \frac{j}{2\pi} \cdot (2\pi j \cdot e^{-2t} \cdot u(t) - 2\pi j \cdot e^{-3t} \cdot u(t))$$

$$= \frac{2\pi \cdot j^2}{2\pi} (e^{-2t} - e^{-3t}) \cdot u(t)$$

$$= (e^{-3t} - e^{-2t}) \cdot u(t)$$

